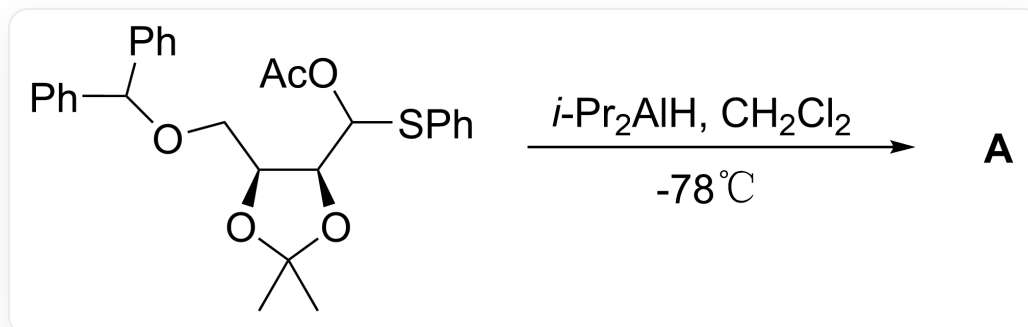
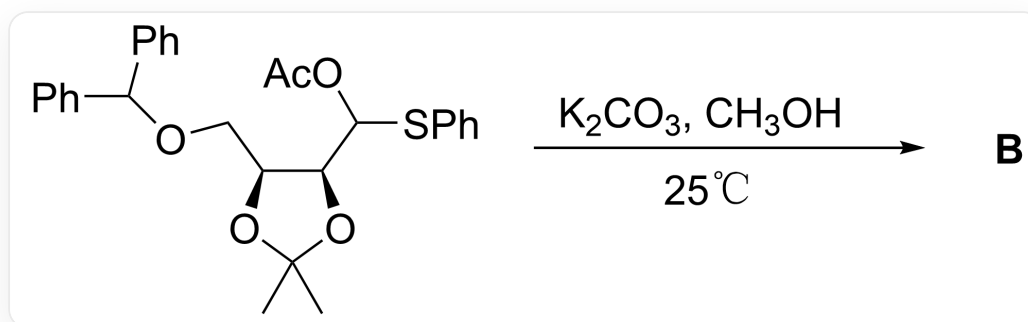


Question



CC1(C)O[C@@H](COC(C2=CC=CC=C2)C3=CC=CC=C3)[C@@H](C(OC(C)=O)SC4=CC=CC=C4)O1>CC(C)[Al]([H])C(C)C.ClCCl, -78°C >[**A**], **A** is the product

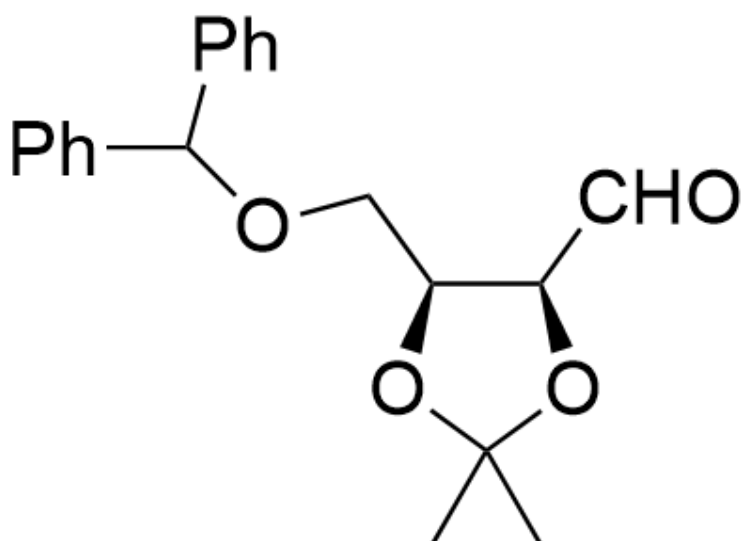


CC1(C)O[C@@H](COC(C2=CC=CC=C2)C3=CC=CC=C3)[C@@H](C(OC(C)=O)SC4=CC=CC=C4)O1>K₂CO₃, CH₃OH, 25°C >[**B**], **B** is the product

Given that the molecular formulas of reaction products **A** and **B** are both $\text{C}_{20}\text{H}_{22}\text{O}_4$, considering stereoisomers, predict the structural formulas of reaction products **A** and **B** respectively.

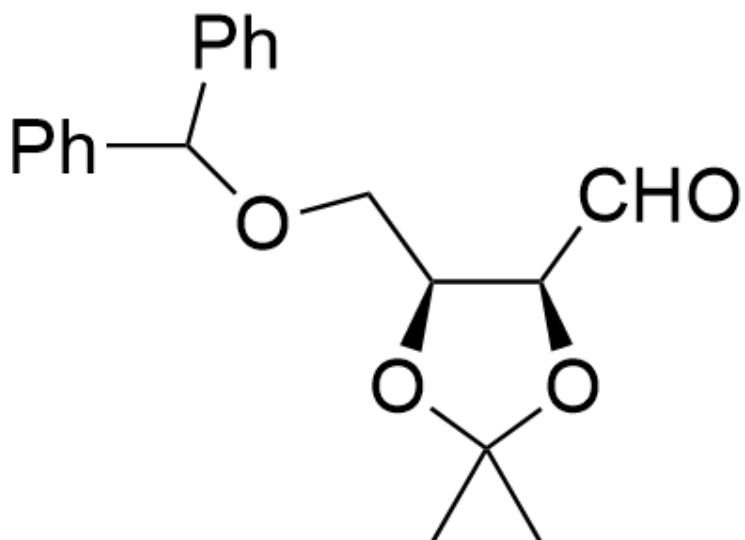
A. All other options are incorrect

B.

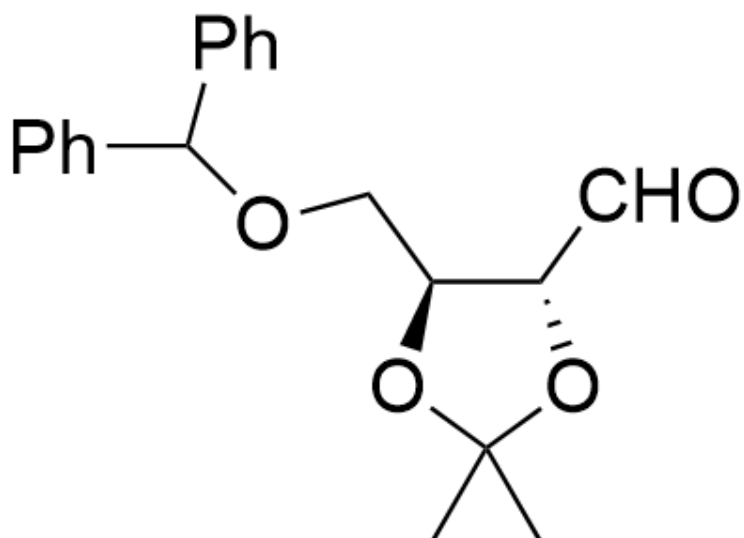


CC1(C)O[C@@H](COC(C2=CC=CC=C2)C3=CC=CC=C3)[C@@H](C=O)O1

C.

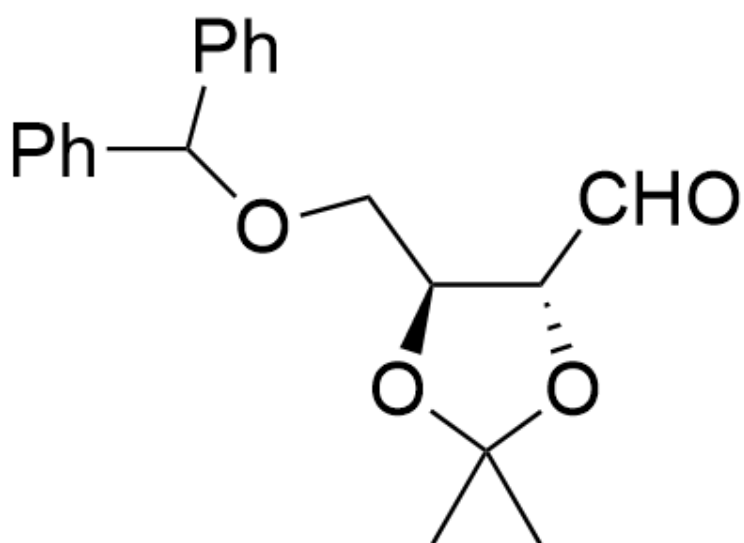


CC1(C)O[C@@H](COC(C2=CC=CC=C2)C3=CC=CC=C3)[C@@H](C=O)O1

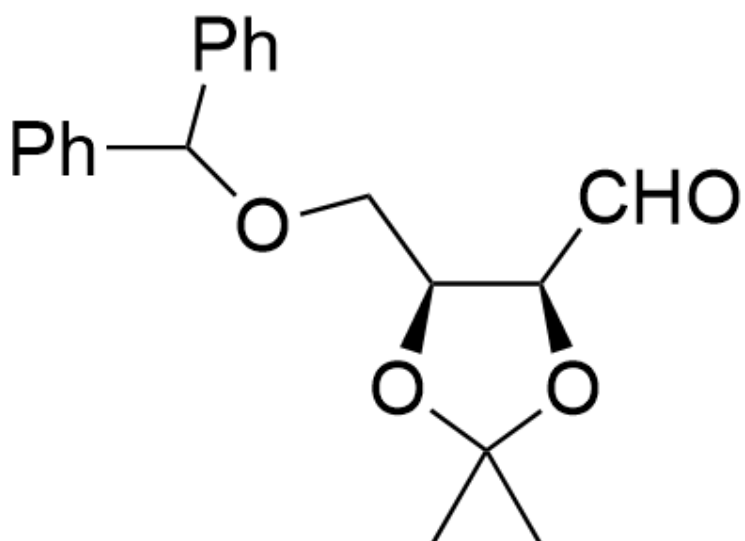


CC1(C)O[C@@H](COC(C2=CC=CC=C2)C3=CC=CC=C3)[C@H](C=O)O1

D.

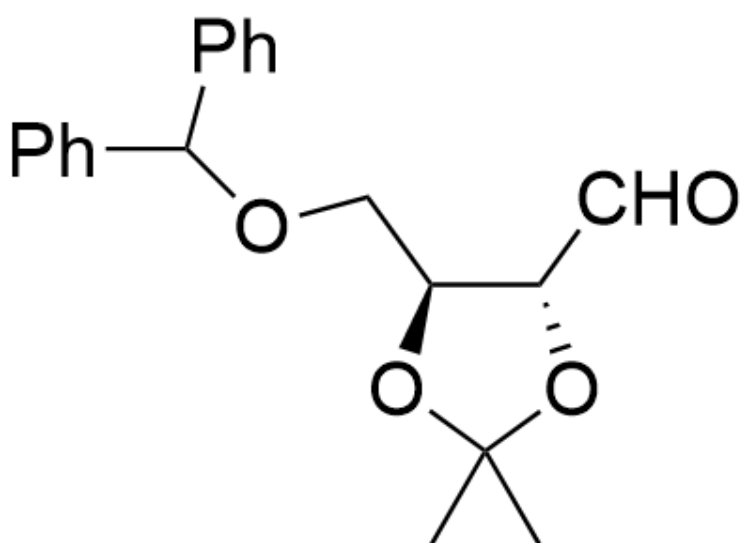


CC1(C)O[C@@H](COC(C2=CC=CC=C2)C3=CC=CC=C3)[C@H](C=O)O1

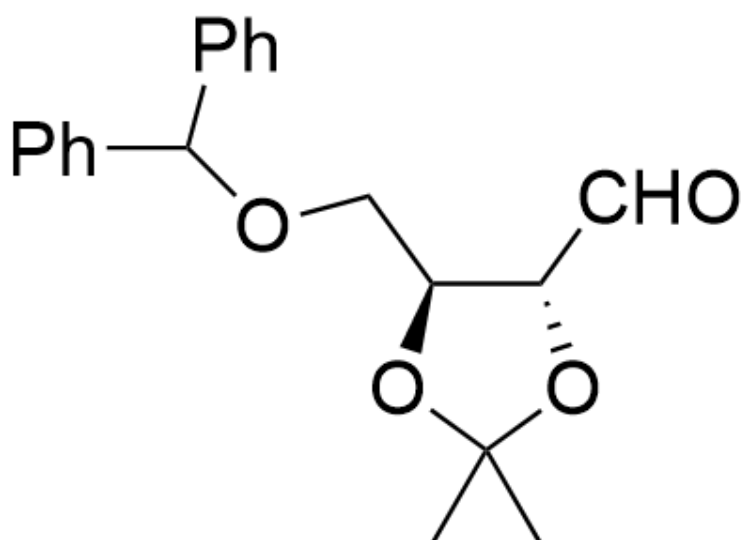


CC1(C)O[C@@H](COC(C2=CC=CC=C2)C3=CC=CC=C3)[C@@H](C=O)O1

E.



CC1(C)O[C@@H](COC(C2=CC=CC=C2)C3=CC=CC=C3)[C@H](C=O)O1



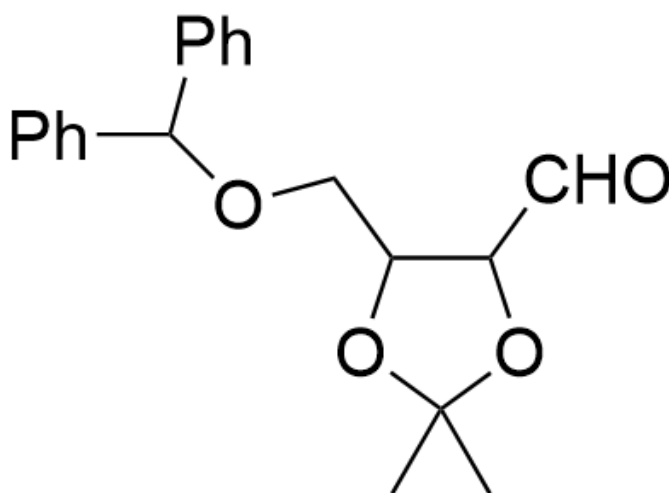
CC1(C)O[C@@H](COC(C2=CC=CC=C2)C3=CC=CC=C3)[C@H](C=O)O1

Answer

Correct Answer: **C**

Detailed Explanation

Based on the molecular formula $C_{20}H_{22}O_4$ of the reaction products **A** and **B**, it can be inferred that the reaction is apparently the deprotection of the aldehyde group, yielding product **1**.



Product **1**: CC1(C)OC(COC(C2=CC=CC=C2)C3=CC=CC=C3)C(C=O)O1

CHECKPOINT

1 PTS

Product **1**: CC1(C)OC(COC(C2=CC=CC=C2)C3=CC=CC=C3)C(C=O)O1

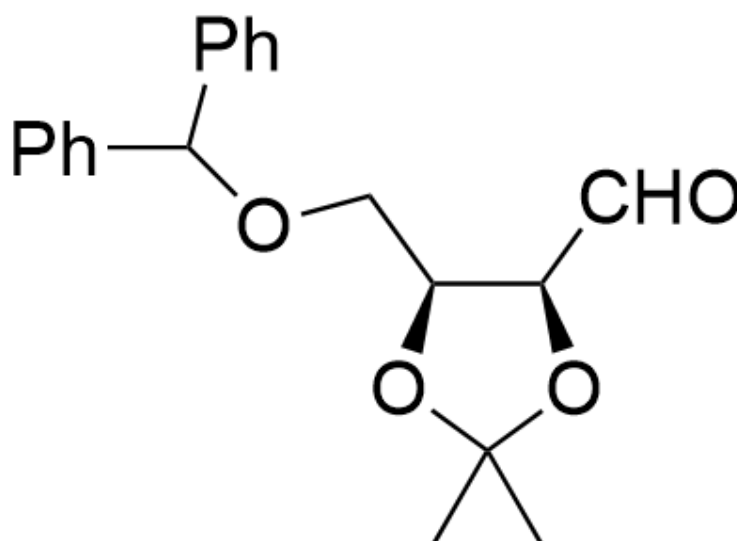
Next, consider stereoselectivity.

The first reaction is carried out in an aprotic solvent at low temperature, so no isomerization occurs after the protecting group is removed, and product **A** with retained configuration is obtained.

CHECKPOINT

1 PTS

The first reaction is carried out in an aprotic solvent at low temperature, so no isomerization occurs after the protecting group is removed, and product **A** with retained configuration is obtained



Product **A**: CC1(C)O[C@@H](COC(C2=CC=CC=C2)C3=CC=CC=C3)[C@@H](C=O)O1

CHECKPOINT

1 PTS

Product **A**: CC1(C)O[C@@H](COC(C2=CC=CC=C2)C3=CC=CC=C3)[C@@H](C=O)O1

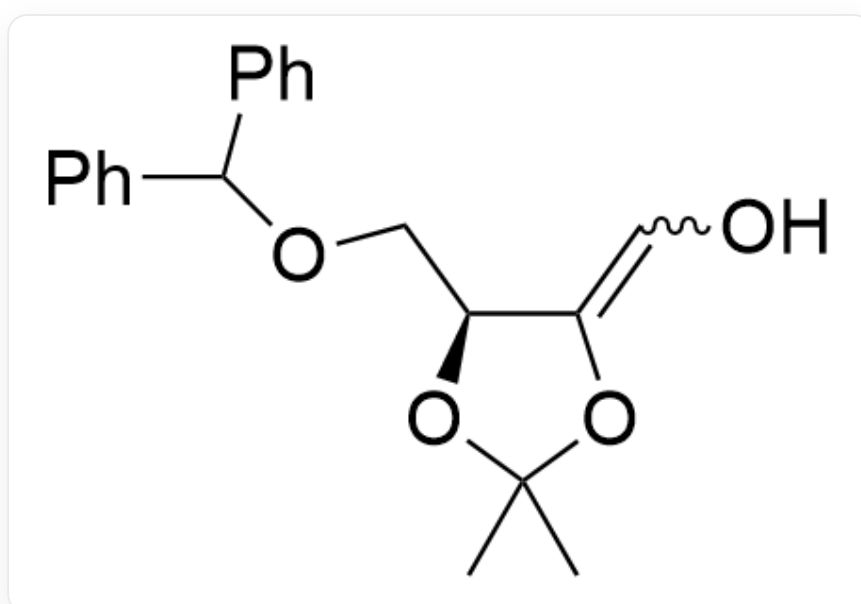
The second reaction is carried out in a protic solvent at room temperature, which facilitates proton transfer reactions. At this time, the epimerization of the aldehyde group at the α position can occur through the enol

intermediate **2**.

CHECKPOINT

1 PTS

The second reaction is carried out in a protic solvent at room temperature, which facilitates proton transfer reactions. At this time, the epimerization of the aldehyde group at the α position can occur through the enol intermediate **2**



Enol intermediate **2**: CC(O/1)(C)O[C@@H](COC(C2=CC=CC=C2)C3=CC=CC=C3)C1=C\O

CHECKPOINT

1 PTS

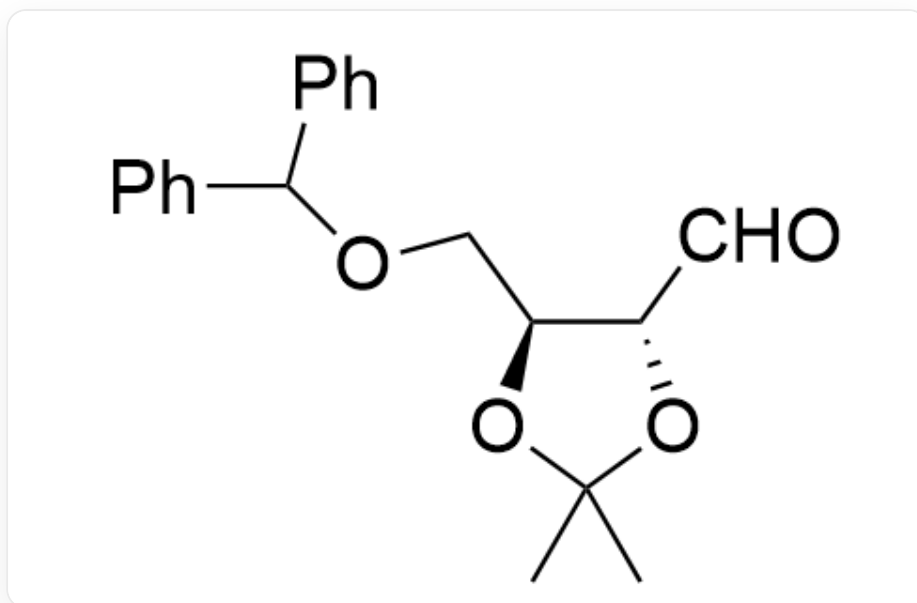
Enol intermediate **2**: CC(O/1)(C)O[C@@H](COC(C2=CC=CC=C2)C3=CC=CC=C3)C1=C\O

Subsequently, the thermodynamically stable product **B** with a greater distance between the aldehyde group and $-\text{CH}_2\text{OCHPh}_2$ is obtained.

CHECKPOINT

1 PTS

The thermodynamically stable product **B** with a greater distance between the aldehyde group and $-\text{CH}_2\text{OCHPh}_2$ is obtained



Product **B**: CC1(C)O[C@@H](COC(C2=CC=CC=C2)C3=CC=CC=C3)[C@H](C=O)O1

CHECKPOINT

1 PTS

Product **B**: CC1(C)O[C@@H](COC(C2=CC=CC=C2)C3=CC=CC=C3)[C@H](C=O)O1